Thermal radiation and color

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use peak wavelength as one piece of evidence about a thermal spectrum.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 13.*
20° from the normal, on the other side.

02. Worked example: explain the model
Hotter curve peaks at shorter wavelength; visible color cannot be inferred from one peak alone without the full spectrum/observer context.

03. Read and interpret the model
6000 K; no, broad spectra.

04. Apply the physics
Hotter peak about half; 970 nm infrared outside visible.

05. Explain your reasoning
Broad thermal spectrum still has visible red tail; peak need not be in visible band.

06. Review: connect an earlier idea *Review: Unit 6, Day 4.*
Upright. An ideal free-end reflection does not invert the displacement pulse.

07. Exit check
Halves.

08. STRETCH: test the model
Peak wavelength halves; much radiation is infrared.

Thermal radiation and color

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use peak wavelength as one piece of evidence about a thermal spectrum.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 13.*
Image moves 0.3 m toward the mirror from behind it. Separation decreases by 0.6 m.

02. Worked example: explain the model
Hotter curve peaks at shorter wavelength; visible color cannot be inferred from one peak alone without the full spectrum/observer context.

03. Read and interpret the model
6000 K; no, broad spectra.

04. Apply the physics
580 nm; 290 nm(UV). Observed color uses whole spectrum and may include emission lines/atmospheric effects.

05. Explain your reasoning
Broad thermal spectrum still has visible red tail; peak need not be in visible band.

06. Review: connect an earlier idea *Review: Unit 6, Day 4.*
First $\lambda=6$ m; second $\lambda=8$ m, using $\lambda=v/f$.

07. Exit check
4000 K.

08. STRETCH: test the model
≈967 and 483 nm; broad emission, visual sensitivity and conditions determine appearance, not one spectral wavelength.